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Team Members

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Introduction to FleetPulse

Current Scenario

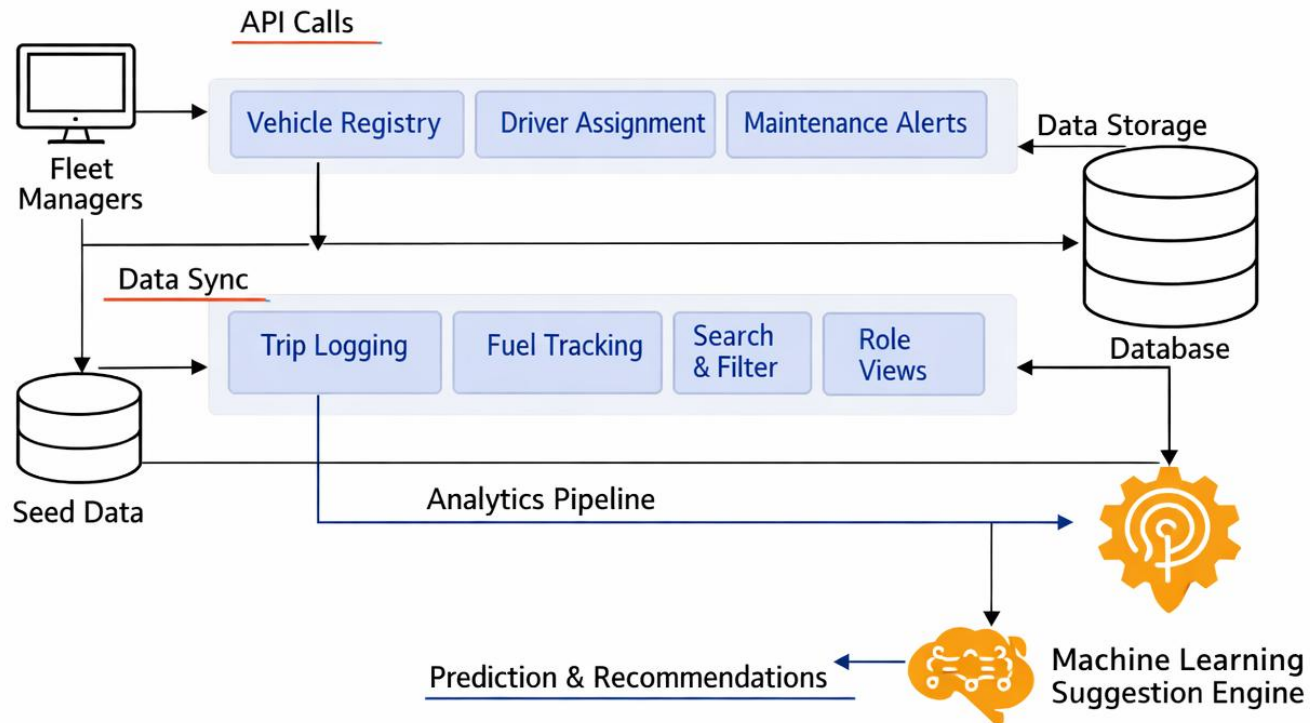
- A regional logistics company manages 120 vehicles and 80 drivers on spreadsheets and phone calls.
- Current inefficiencies include:
 - - Missed maintenance schedules
 - - Double-assigned drivers
 - - Poor visibility into fleet utilization and costs
-

Approach:

Build a web application to automate the process and proper fleet management. Automation ensures accuracy, transparency, and efficiency across fleet operations. This shifts fleet management from manual oversight to automated intelligence, enabling faster decisions, reduced downtime, and optimized resource use.

Design

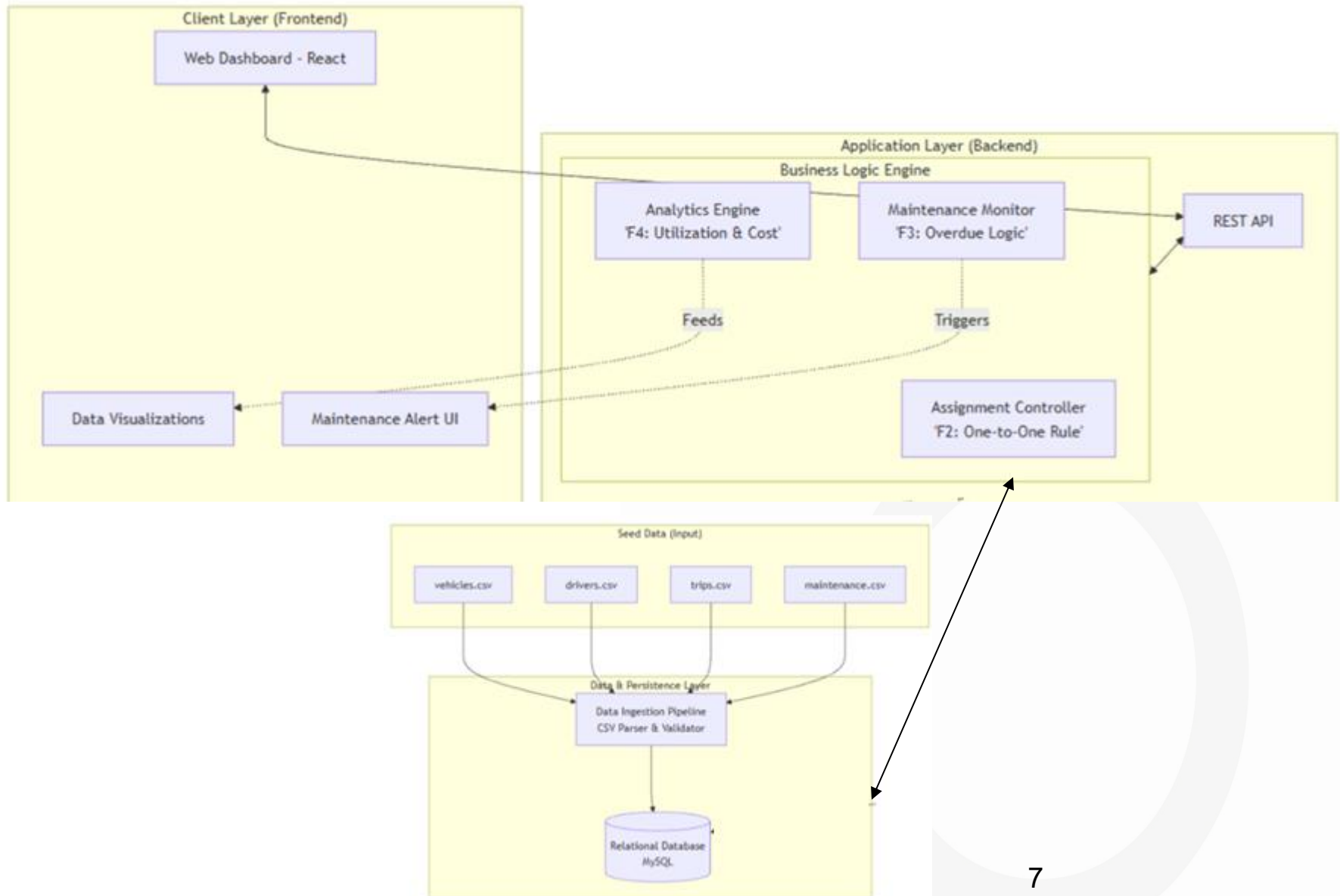
FleetPulse Architecture Diagram



Technology Used

- Front End: ReactJS
- Web API: NodeJS
- Database: MySQL.
- Version controller: GitHub.
- Development methodology : Agile & Scrum.

Process Flow



Functionality -1

Vehicle registry

- List all vehicles with status (active / in shop / idle)
- detail view per vehicle from seed data. CRUD operations

Driver roster & assignment

- View drivers
- assign a driver to a vehicle
 - Check: system prevents assigning one driver to two vehicles.

Maintenance alerts

- Vehicles overdue by mileage or service date are flagged visibly (rules in starter kit README). Show in notification icon with list and also in menu.

KPI dashboard

- Total vs active vehicles
- fleet utilization %
- overdue-maintenance count
- top 5 highest-cost vehicles.

Functionality -2

Trip logging & history

- Record a trip (vehicle, driver, distance, date)
- filterable history view

Fuel & cost tracking

- Fuel/cost entries per vehicle
- simple monthly cost report

Search & filter

- Search registry by plate, model, status, assigned driver.

Role views

- Dispatcher view (assignments, trips) vs fleet-manager view (KPIs, costs).

Functionality -3

Maintenance risk score

- Per-vehicle risk score from mileage, age, history.
- A transparent heuristic is fine.

Natural-language fleet Q&A

- e.g. "Which vehicles need service this week?" answered from your data.

Assignment suggestions

- Suggest driver–vehicle assignments (availability, utilization balance).

FleetPulse Business Rules

-Maintenance Overdue Rule

A vehicle is OVERDUE for service if either:

More than 10,000 km since its last service

'odometerkm-lastserviceodometerkm>10000'

More than 180 days since last_service_date

A vehicle with no service history is overdue by definition.

-Vehicle Assignment Rule

A driver may be assigned to at most one vehicle.

A vehicle may be assigned to at most one driver.

Only active drivers and active vehicles can be newly assigned.

The system must prevent double-assignment.

-Fleet Utilization

Defined as:

Fleet Utilization % = {Active Vehicles with Assigned Driver}/{Total Vehicles}*100

FleetPulse Business Rules

Continued...

-Top Cost Vehicles

Rank vehicles by total operating cost over the trip data period:

fuel_cost (from trips), cost (from maintenance)

-Integrity Guarantees

All vehicle_id and driver_id references in trips and maintenance must resolve.

No driver is assigned to two vehicles; no vehicle has two drivers.

Assignment columns in the two files must agree.

Trips must only exist for driver–vehicle pairs that are assigned to each other.

-All other cases — empty fields, edge cases, odd distributions — are intentional and must be tested against.

Conclusion

Automation transforms FleetPulse from a **manual monitoring tool** into a **command center** that:

- Reduces downtime
- Cuts costs
- Improves safety
- Provides actionable intelligence for strategic fleet management

Additionally:

Maintenance reliability improves: Vehicles overdue for service are automatically flagged using mileage and date thresholds, ensuring proactive scheduling and reducing breakdown risk.

Driver–vehicle integrity is guaranteed: Double-assignments are prevented, and only active drivers/vehicles can be paired, eliminating operational conflicts.

Fleet utilization becomes transparent: A clear percentage metric shows how effectively the fleet is deployed, enabling managers to spot idle capacity and optimize usage.

Cost hotspots are visible: Automated ranking of the top 5 highest-cost vehicles highlights inefficiencies, guiding decisions on repairs, replacements, or reassignment.

Data integrity is enforced: Automated checks ensure all IDs resolve, assignments align, and trips exist only for valid driver–vehicle pairs, strengthening trust in analytics.

Decision-making accelerates: With KPIs, alerts, and risk scores surfaced in real time, managers move from reactive firefighting to proactive fleet optimization.

Future enhancements

Prediction & Recommendations

In the FleetPulse architecture, the “**Prediction & Recommendations**” flow represents the intelligence layer powered by the machine learning engine. It typically includes outputs that help optimize fleet operations and decision-making.

Here’s what it can include:

-Predictive Insights

Maintenance forecasting: Predict which vehicles are likely to need service soon based on mileage, age, and usage patterns.

Fuel consumption trends: Estimate future fuel costs and identify inefficiencies.

Driver performance risk: Flag drivers with higher likelihood of incidents or delays.

-Operational Recommendations

Driver–vehicle assignment suggestions: Match drivers to vehicles based on availability, utilization balance, and maintenance schedules.

Route optimization: Recommend efficient routes to reduce travel time and fuel usage.

Fleet utilization improvement: Suggest redistributing idle vehicles or rescheduling trips for better resource use.

-Managerial Alerts

Overdue maintenance prioritization: Rank vehicles by urgency for service.

Cost optimization tips: Highlight vehicles with high operating costs and suggest corrective actions.

Anomaly detection: Detect unusual trip patterns or cost spikes.

Essentially, this layer turns raw data into actionable intelligence — helping fleet managers make proactive, data-driven decisions.

Future enhancements

Continued...

Machine Learning Suggestion Engine

The Machine Learning Suggestion Engine is the intelligence core that transforms operational data into smart, automated insights. It's where algorithms analyze historical and real-time fleet data to make predictions and recommendations that improve efficiency, safety, and cost management.

Here's what it typically means and includes:

-Core Purpose

Analyze patterns in vehicle usage, maintenance, fuel consumption, and driver behavior.

Generate predictions (e.g., when a vehicle will need service, which routes minimize fuel cost).

Provide recommendations to fleet managers for better decision-making.

THANK YOU

